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BREWINGSEC
CYBER COMMUNITY

DEEPFAKE PHISHING: HOW AI-POWERED CONTENT IS RESHAPING SOCIAL ENGINEERING

BY ABHIJEET

DISCLAIMER

The techniques shown in this talk are offensive in nature

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Neither me nor the event organizers will be liable for any misuse of the talk materials.

WHOAMI

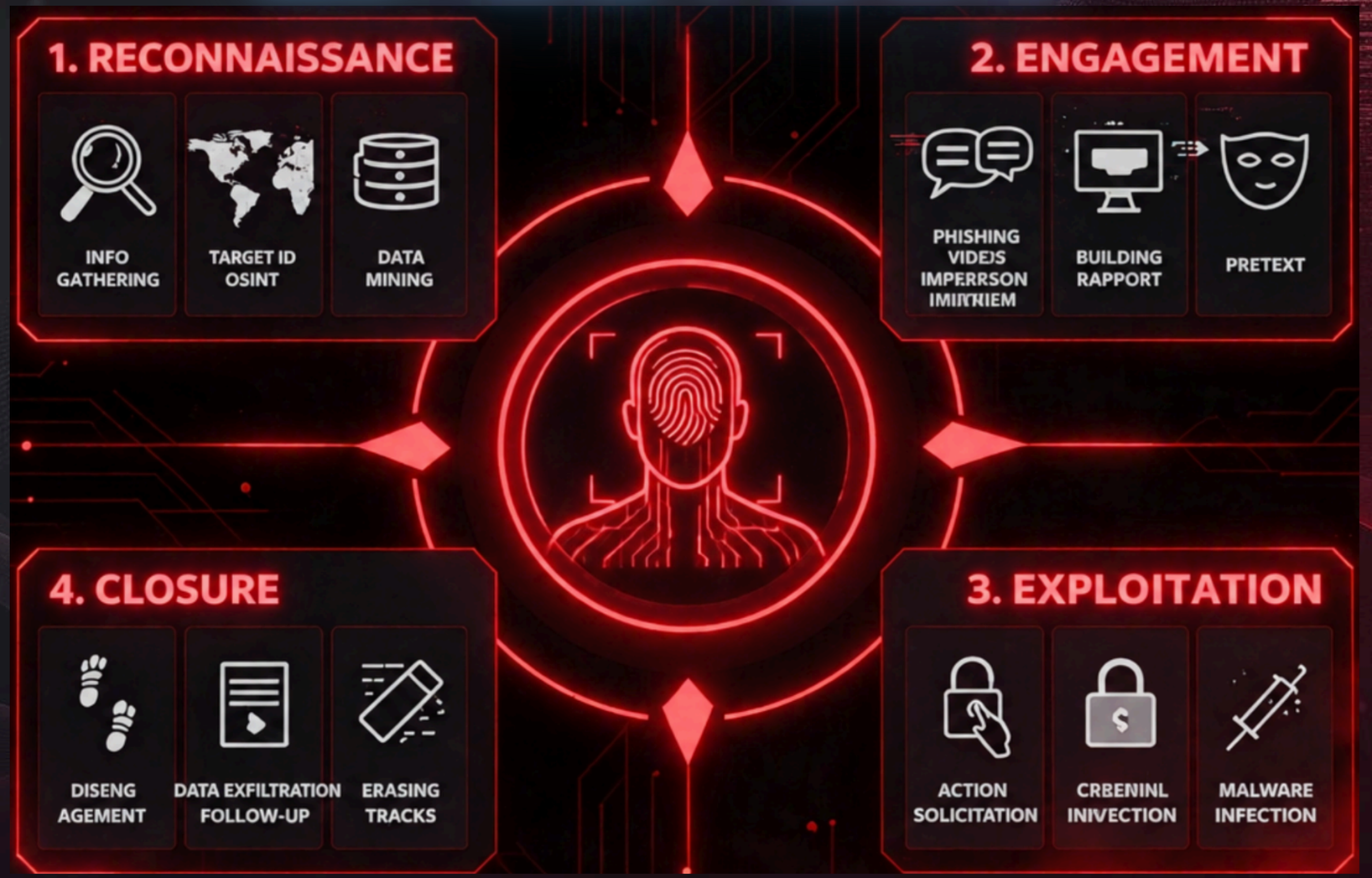
Security Researcher & Tinkerer

Research areas include Adversarial TTPs, *NIX environments, cloud infrastructure, and supply chain pipelines

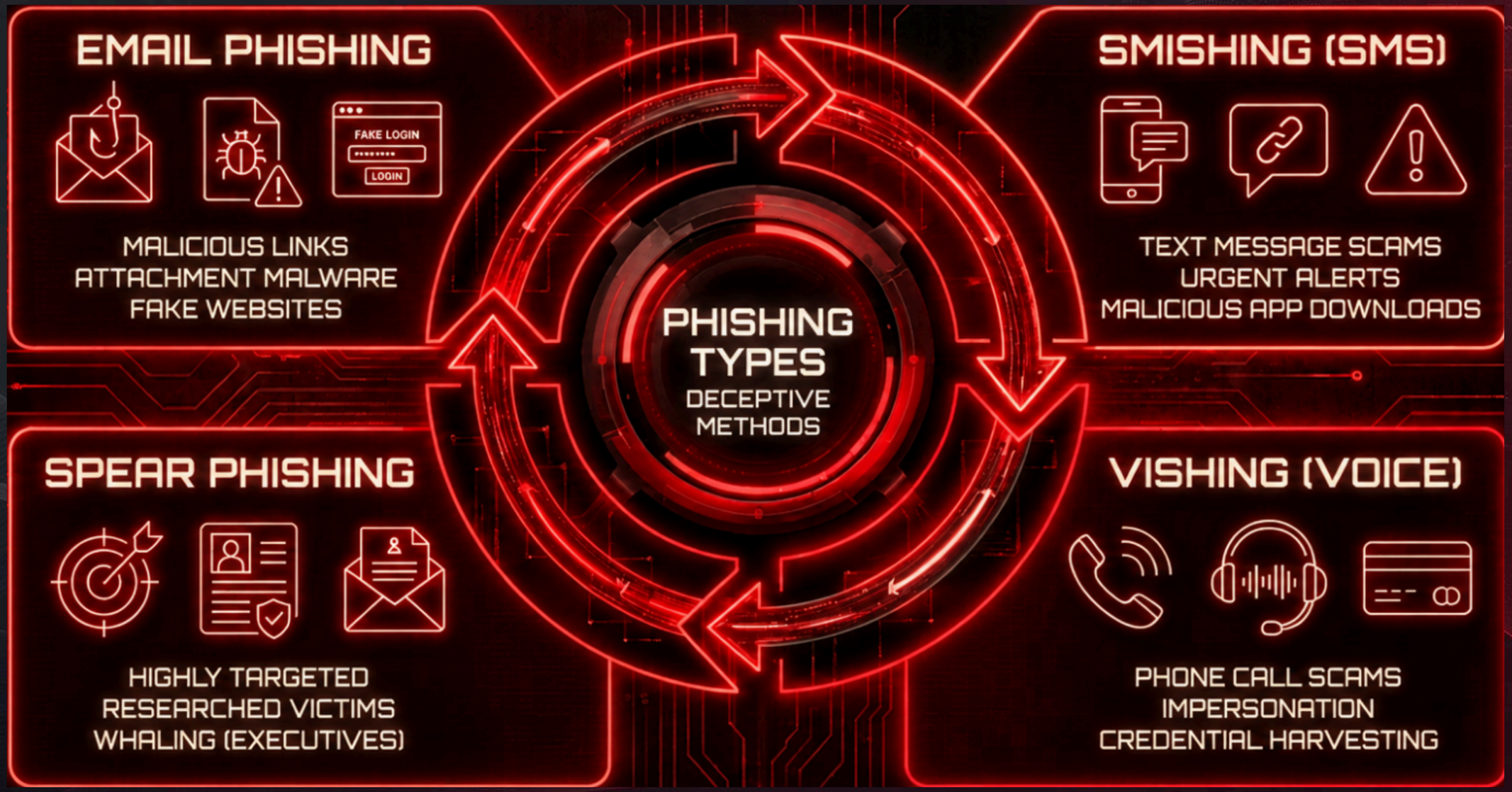
Full-time human, part-time homechef

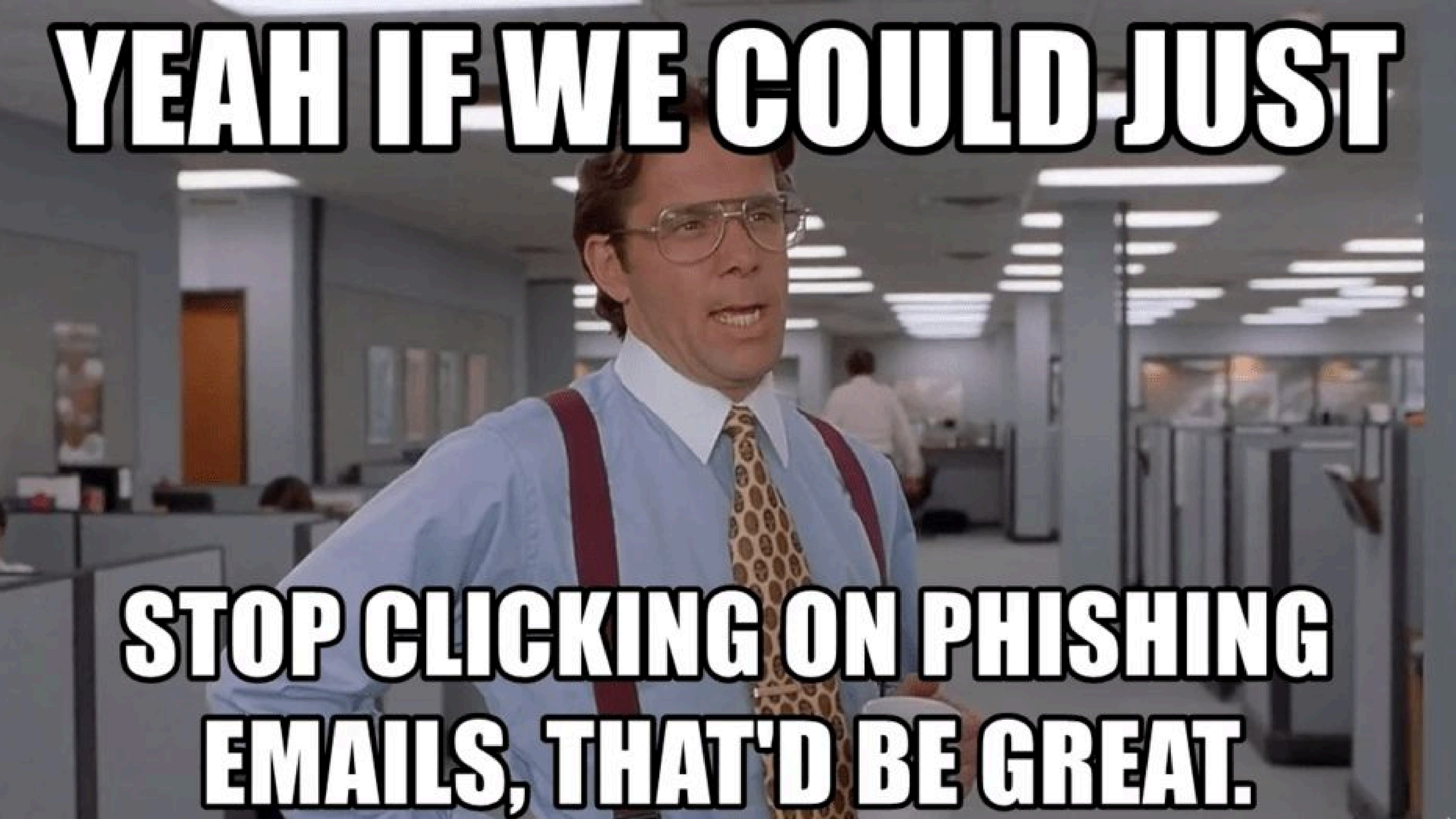
Current gig :- Cloud Threat Research @ Qualys

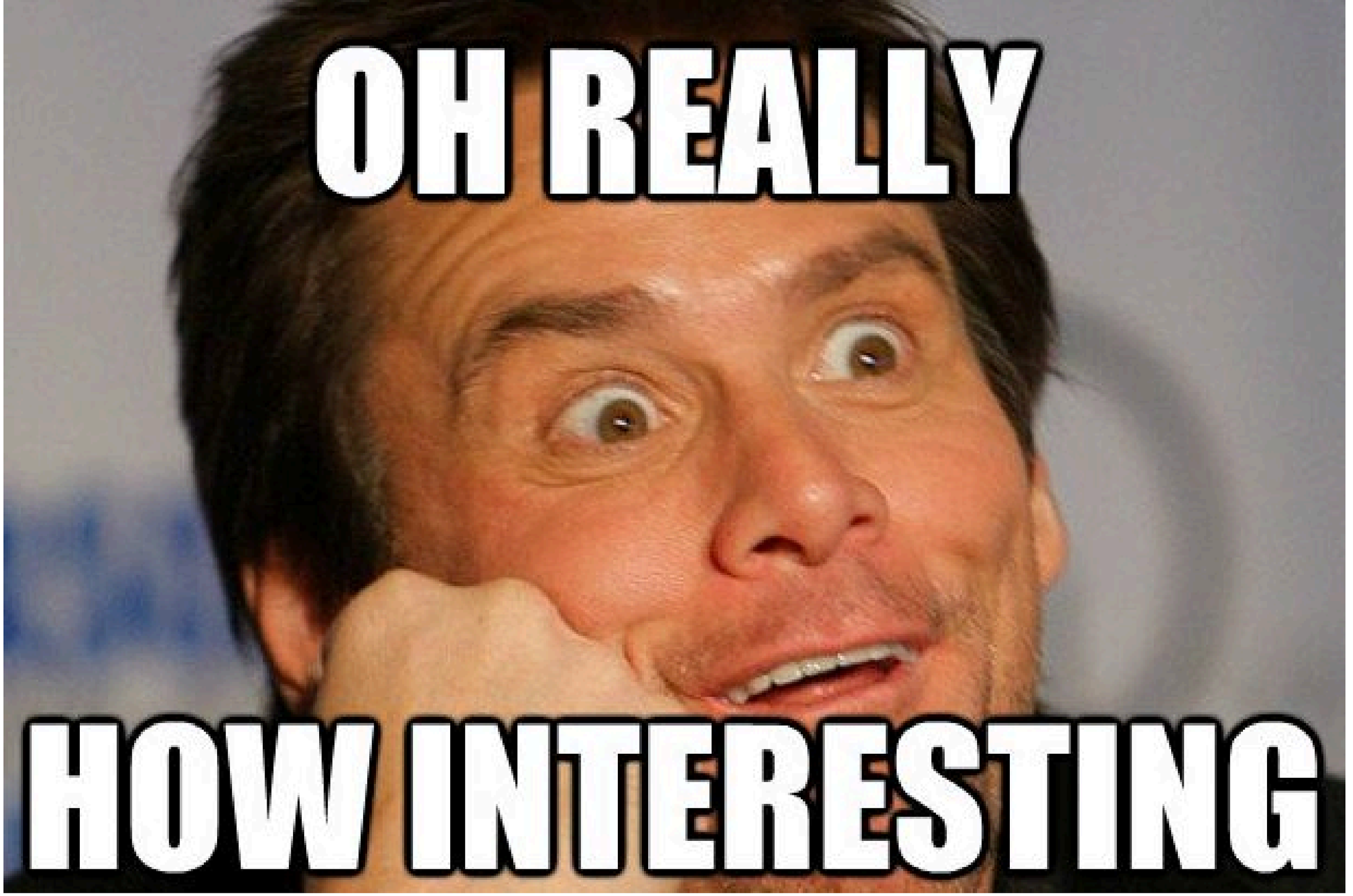
Social Engineering Lifecycle



PHISHING TYPES




YEAH IF WE COULD JUST
STOP CLICKING ON PHISHING EMAILS, THAT'D BE GREAT.


OH REALLY
HOW INTERESTING

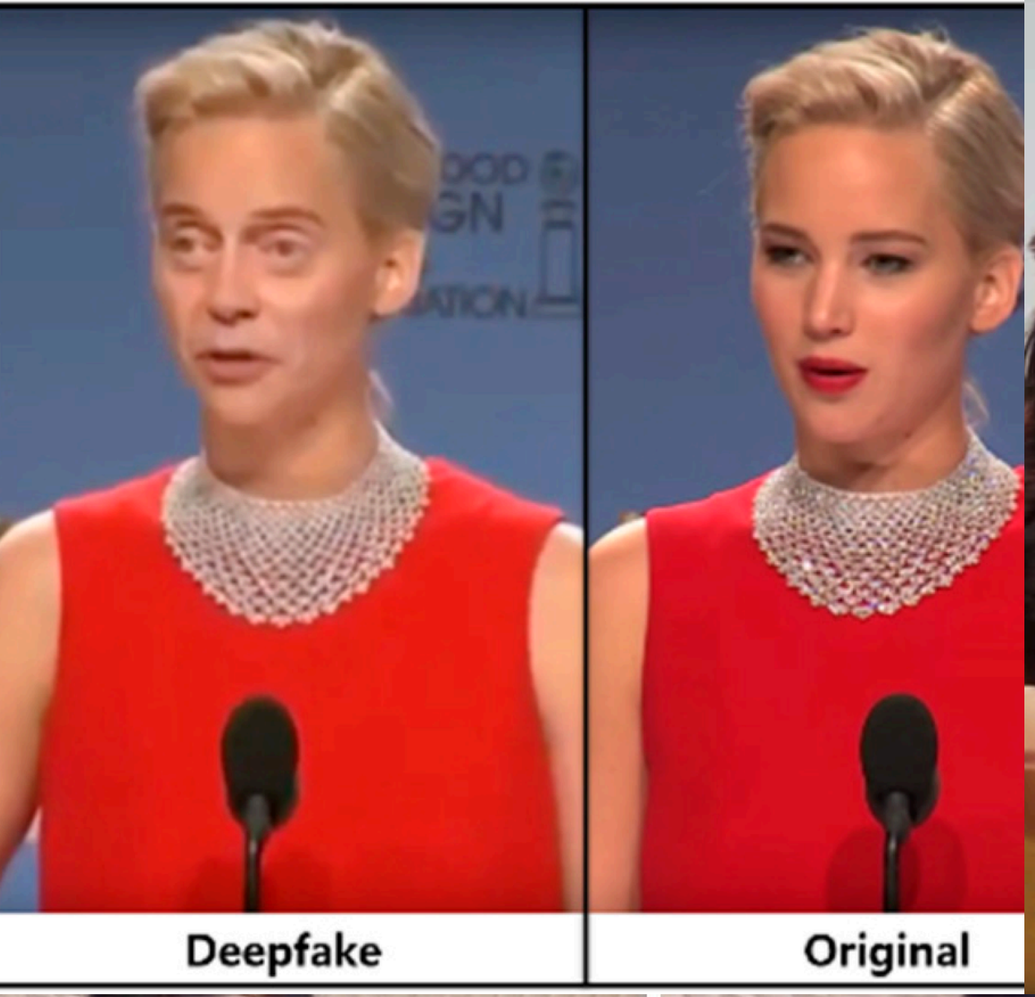
meme-arsenal.ru

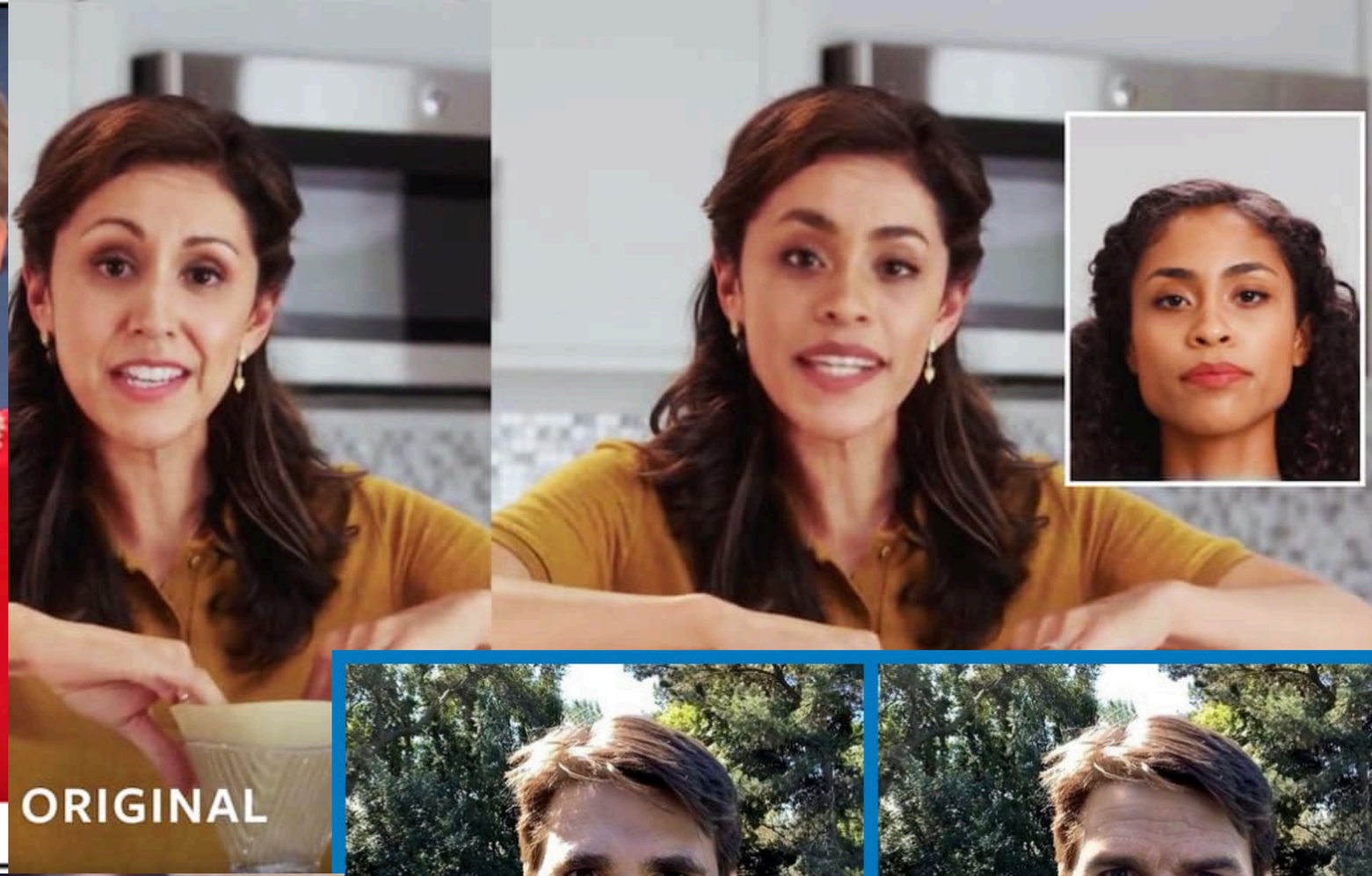
DEEP
FAKE

NOW WHAT

THE BLEEP IS

A DEEPFAKE?


Deepfake
Original


ORIGINAL

WHERE DO THEY COME FROM?

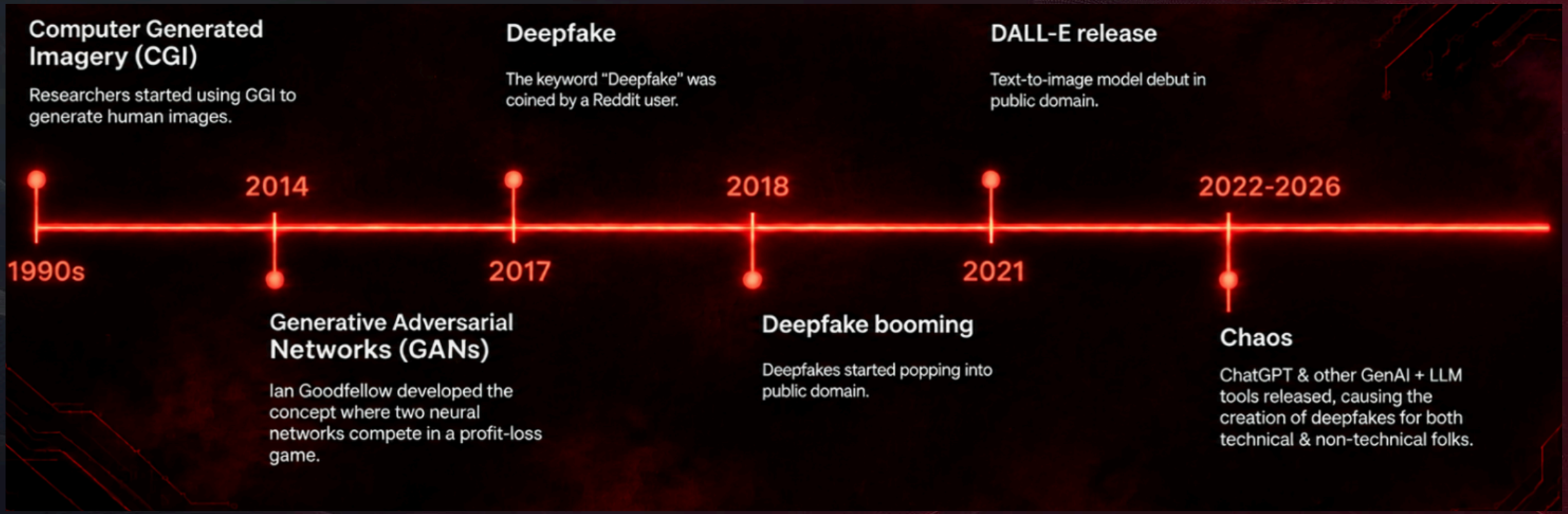
You get a LLM

LLM 101

- LLMs (Large Language Models) are subclass of artificial neural networks trained on massive datasets.
- Advanced LLMs comes under Generative Pre-trained Transformers (GPT).
- These models can write emails, create conversations, and mimic writing styles with remarkable accuracy.

You must be new here

Deepfake History



Generative Adversarial Networks

Generator

The Generator creates synthetic content from random noise, attempting to produce outputs indistinguishable from real data. It learns to map latent space to realistic samples through iterative training.

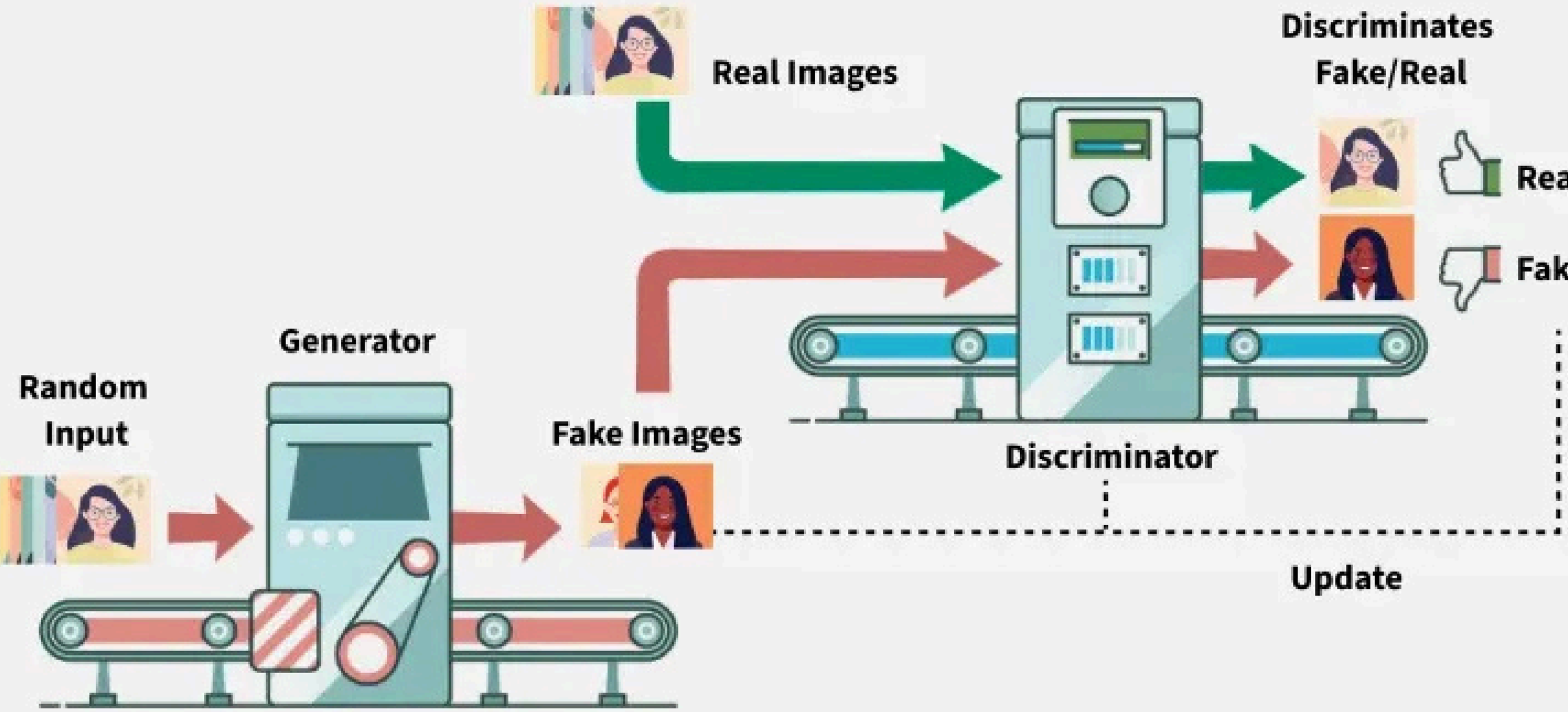
Discriminator

The Discriminator acts as a critic, evaluating whether inputs are real or generated. It provides feedback that helps the Generator improve, creating an adversarial training loop.

Adversarial Loop

Both networks compete in a zero-sum game. As the Generator improves at creating fakes, the Discriminator gets better at detection, driving increasingly realistic outputs.

Generative Adversarial Network (GANs)



DEEPFAKE TYPES

FACE SWAP

Replaces one person's face with another in video or images. Uses encoder-decoder neural networks to map facial features and expressions seamlessly onto a target face.

LIP SYNC

Manipulates mouth movements to match different audio. AI analyzes speech patterns and generates realistic lip movements synchronized with fabricated or altered audio tracks.

FULL BODY

Generates complete synthetic human figures with realistic movements and gestures. Combines pose estimation with generative models to create convincing full-body puppetry.

VOICE CLONE

Synthesizes realistic voice replicas from audio samples. Neural networks learn vocal characteristics, tone, and speech patterns to generate convincing synthetic speech.

AI-Powered Social Engineering

Automated Targeting

AI algorithms analyze vast amounts of public data from social media, corporate websites, and leaked databases to identify high-value targets. Machine learning models profile victims based on behavior patterns, job roles, and network connections to maximize attack success rates.

Personalization

Large language models generate hyper-personalized phishing messages that reference specific projects, colleagues, and recent activities. These AI-crafted communications bypass traditional detection by mimicking authentic writing styles and contextual relevance.

Scale & Speed

AI enables attackers to launch thousands of unique, tailored attacks simultaneously. Automated systems adapt in real-time based on victim responses, creating dynamic conversation flows that maintain engagement and increase conversion rates dramatically.

GENERATE YOUR OWN DEEPFAKES (GYOD)

Voice Cloning

1. OmniVoice (HuggingFace)
2. ElevenLabs (Paid)
3. Respeecher (Paid)

Video Deepfakes

1. DeepFaceLab
2. DeepFaceLive
3. FaceSwap

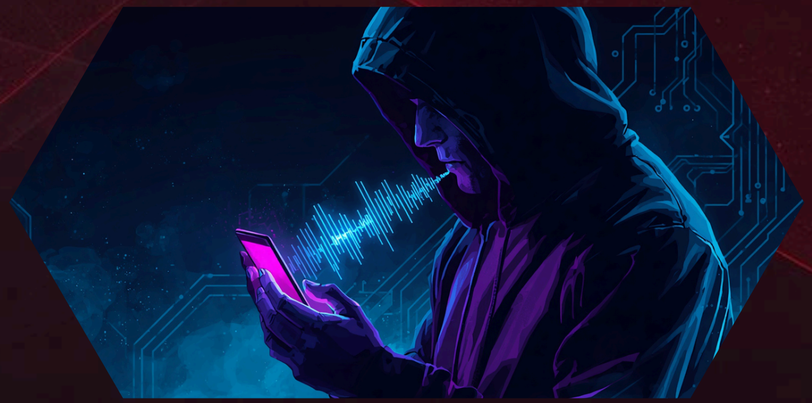
Case Study 01: Video Deepfake

DEMO 01:

VIDEO

DEEPFAKE

CASE STUDY 02: VOICE CLONING



DEMO 02:

VOICE CLONING

Deepfake Detection Methods

Visual Artifacts

Look for unnatural blinking patterns, inconsistent lighting on faces, blurry edges around hairlines, and distorted facial features during movement. Pay attention to asymmetrical faces and misaligned teeth or ears.

Audio Analysis

Detect robotic or unnatural speech patterns, inconsistent background noise, and audio-visual sync issues. Voice cloning often produces subtle pitch anomalies and breathing pattern irregularities.

Metadata Review

Examine file metadata for editing software signatures, creation timestamps, and compression artifacts. Authentic media typically has consistent metadata trails while deepfakes show manipulation indicators.

AI Detection Tools

Leverage specialized AI tools which analyze pixel-level inconsistencies and neural network fingerprints invisible to human eyes.

Deepfake Detection Tools

1. Awesome Deepfake Detection: https://github.com/Daisy-Zhang/Awesome-Deepfakes-Detection
2. Deep-Fake-Detector-v2-Model: https://huggingface.co/prithivMLmods/Deep-Fake-Detector-v2-Model
3. FaceForensics: https://github.com/ondyari/FaceForensics

Key Takeaways

Awareness

Defense

Q&A

THANK YOU